

IIT JODHPUR

OS Security & Permissions

WEEK 8 · LIVE PRACTICAL SESSION

IIT Jodhpur — BS/BSc Programme · Operating Systems · Semester 3

Today you'll think like both a **system admin** and a **security analyst** — reading permissions, locking down files, and mapping attack surfaces inside a real Linux terminal. Let's get hands-on. 🔒

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Session Agenda — 120 Minutes

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Here's exactly how our 2-hour session is structured. Slides are just the launchpad — the real action happens in the terminal.

0–30 min 1

🧠 Revision — CIA Triad, AAA Model & Access Control Concepts

2 **30–70 min**

🖥️ Activities 1 & 2 — File Permissions Explorer + chmod/chown Manager

70–85 min 3

🐍 Activity 3 — CIA Triad Visualizer in Google Colab (Python)

4 **85–115 min**

📝 Assignments + Pitfalls Review

115–120 min 5

🎟️ Exit Ticket, Homework & Q&A

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TOOLS SETUP · 5–10 MIN

🛠️ Tool Setup — Before We Begin

Tool 1 — copy.sh/v86 Buildroot

Open Chrome → go to **copy.sh/v86**

Click **'Images'** dropdown (top left)

Select **'Buildroot'**

Click green **'Start emulation'** button

Wait 30–60 sec for **/ #** prompt → You're root!

⚠ ⚠ copy.sh v2 is DEAD. Always use copy.sh/v86 with Buildroot only.

Tool 2 — Google Colab

Go to **colab.research.google.com**

2. Sign in with your Google account

Click **'New Notebook'** (blue button)

4. Click inside the empty code cell

Press **Shift + Enter** to run a cell

✅ Both tools run in your browser — no installation needed!

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REVISION · 10–20 MIN

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🏏 CIA Triad — The IPL Dressing Room Analogy



🔒 Confidentiality

The captain's match strategy doc. **Only coaches and captain can read it.***WHO CAN SEE?*

✅ Integrity

The scoreboard. **Only the official scorer can** *MODIFY***change it.***WHO CAN ?*

📡 Availability

The stadium scoreboard **must be visible to all fans at all times.***IS IT USABLE?*

🪪 Authentication

Players show ID cards to enter.

WHO ARE YOU?

🚪 Authorization

A player can enter the dressing room, **NOT the BCCI finance office.***WHAT CAN YOU DO?*

📹 Accountability

CCTV records who entered which room and when.

WHO DID WHAT?

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REVISION · 20–30 MIN

Access Control Models — DAC, MAC & RBAC

The professor explained three models in the lecture. Here's how they compare side by side:

Model	Who Decides?	Owner Can Override?	Real-World Example
DAC — Discretionary	Resource Owner	✅ YES	Your personal laptop files
MAC — Mandatory	System Admin / Policy	❌ NO	Army / Defence files (Top Secret)
RBAC — Role-Based	Role Assignment	❌ NO (Role-defined)	College ERP — Student vs. Professor

📝 Quiz check: "A professor can upload grades but a student can only view them" → This is **RBAC**. "Even the file owner cannot change permissions" → This is **MAC**.

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ACTIVITY 1 · 30–50 MIN

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🖥️ Activity 1 — File Permissions Explorer

Every Linux file has a permission string like `-rwxr-xr--` — this is **DAC in action**. You'll create files, inspect permissions, and decode each character like a system admin.

Step 1 — Create Your Lab Directory

```
mkdir /home/security_lab
cd /home/security_lab
pwd
```

`mkdir` creates a folder · `cd` moves into it · `pwd` confirms your location → expected: `/home/security_lab`

Step 2 — Create Three Test Files

```
echo 'Aadhaar data for user Rahul'
> secret_data.txt
echo 'Public timetable info' >
public_info.txt
echo '#!/bin/sh' > run_script.sh
echo 'echo Hello from security
lab!' >> run_script.sh
```

`>` creates/overwrites · `>>` appends without overwriting

Step 3 — Inspect Permissions

```
ls -l
```

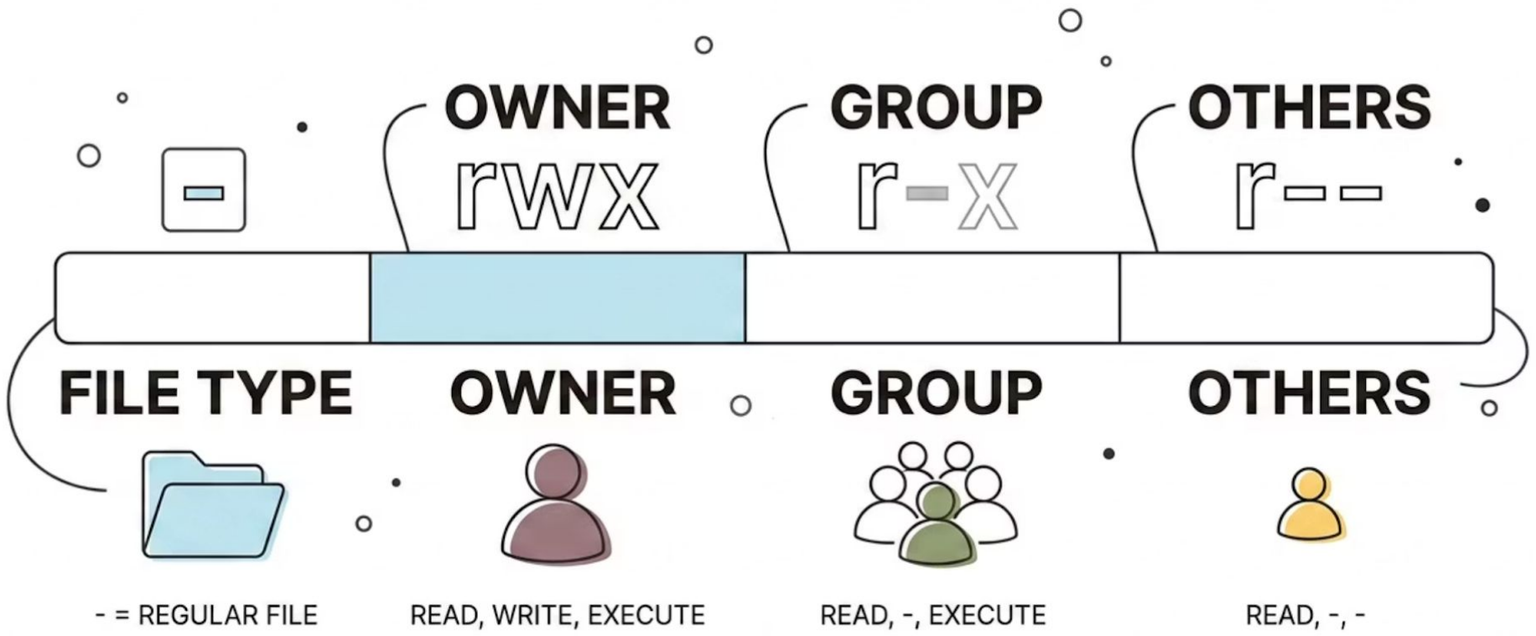
Expected: `-rw-r--r--` for all three files.

Positions 2–4 = owner · 5–7 = group · 8–10 = others.

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🔍 Reading the Permission String

Every `ls -l` output starts with a 10-character permission string. Here's how to decode it:



Owner (positions 2–4)

`rw-` = can Read + Write, cannot Execute

Group (positions 5–7)

`r--` = can only Read; no write, no execute

Others (positions 8–10)

`r--` = can only Read; complete restriction otherwise

Default new file permissions in Linux: `-rw-r--r--` — owner reads/writes, everyone else reads only.

ACTIVITY 2 · 50–70 MIN

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🔑 Activity 2 — chmod & chown Permission Manager

Now we **change** permissions using `chmod`. The octal method uses three digits — one per identity group. Memorize this formula: **r=4, w=2, x=1**. Add them up for each group.

chmod 600

`secret_data.txt`Owner: rw- · Group: --- · Others: ---

Only YOU can read/write. Nobody else. Like Aadhaar data.

chmod 700

`run_script.sh`Owner: rwx · Group: --- · Others: ---

Only owner can run the script. Like a bank's private automation.

chmod 644

`public_info.txt`Owner: rw- · Group: r-- · Others: r--

Everyone reads, only owner edits. Like a notice board.

chmod 777

NEVER use in production!Owner: rwx · Group: rwx · Others: rwx

Like a public toilet with no lock — anyone can do anything.

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chmod in Practice — Scenarios & Commands

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Run These Commands in Your Terminal

```
# Scenario 1: Make secret_data.txt private
chmod 600 secret_data.txt
ls -l secret_data.txt
# Expected: -rw------- root root ... secret_data.txt

# Scenario 2: Make script executable (owner only)
chmod 700 run_script.sh
sh run_script.sh
# Expected output: Hello from security lab!

# Scenario 3: Public notice board file
chmod 644 public_info.txt
ls -l public_info.txt
# Expected: -rw-r--r-- root root ... public_info.txt

# Scenario 4: Inspect user account database
cat /etc/passwd
head -5 /etc/passwd
wc -l /etc/passwd
```

Octal Quick Reference

Octal	Binary	Meaning
7	111	rwx
6	110	rw-
5	101	r-x
4	100	r--
0	000	---

`/etc/passwd` stores ALL user accounts. Each line = one user. `wc -l` counts how many accounts exist on the system.

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ACTIVITY 3 · 70–85 MIN

🐍 Switch to Google Colab — CIA Triad Visualizer

⚠ ⚡ ACTION: Open Google Colab now → colab.research.google.com → New Notebook. Copy each cell below into a separate code cell and run with Shift + Enter.

01

Cell 1 — CIA Triad Simulator

Define a `files` dictionary with `authorized_readers`, `authorized_writers`, and `content`. Implement `check_confidentiality()` and `check_integrity()` functions. Run test cases — observe ALLOWED vs. DENIED results.

02

Cell 2 — Availability Simulation

Use `toggle_availability()` to simulate a file going offline (like a DoS attack) and coming back online. Try reading a file while it's offline — watch it fail gracefully.

03

Cell 3 — Bar Chart Visualization

Use `matplotlib` to draw a bar chart of all CIA test results. Green bars = ALLOWED, Red bars = DENIED. Chart is saved as `cia_triad_results.png`.

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ASSIGNMENT 1 · 85–95 MIN

📝 Assignment 1 — Permission Audit Challenge

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You are a junior sysadmin at **SecureTech**. Your manager has given you three files to lock down using correct permissions. This is real-world DAC in action.

Your Commands

```
cd /home/security_lab

# File 1: Company financial data
echo 'Q3 Revenue: 50 Crore INR' > finance_report.txt

# File 2: HR policy document
echo 'Leave policy: 20 days per year' > hr_policy.txt

# File 3: Backup script
echo '#!/bin/sh' > backup.sh
echo 'echo Running backup...' >> backup.sh

# Apply permissions
chmod 600 finance_report.txt
chmod 644 hr_policy.txt
chmod 700 backup.sh

# Audit — verify all three
ls -l /home/security_lab/

# BONUS: Run the script
sh backup.sh
```

Expected Audit Output

```
-rw------- root root ... finance_report.txt
-rw-r--r-- root root ... hr_policy.txt
-rwx------ root root ... backup.sh
```

Completion Checklist

`finance_report.txt` shows `-rw-------` (600)

`hr_policy.txt` shows `-rw-r--r--` (644)

`backup.sh` shows `-rwx------` (700)

`sh backup.sh` outputs: *Running backup...*

✅ Done? Show your `ls -l` output to the TA before moving on.

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ASSIGNMENT 2 · 95–105 MIN

📝 Assignment 2 — Attack Surface Mapper

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The professor said: **"Network plus services is the most important attack surface."** Investigate your running system like a real security analyst — covering the Identity, Process, and Storage attack layers.

1

Identity Layer — User Accounts

```
cat /etc/passwd
wc -l /etc/passwd
ls -l /etc/passwd
ls -l /etc/shadow
```

Count users, compare permissions of `/etc/passwd` vs. `/etc/shadow`. Why is shadow more restricted?

2

Process Layer — Running Processes

```
ps aux
ps aux | wc -l
ps aux | grep ash
```

The pipe `|` chains commands. `ps aux` → `grep ash` filters for the Buildroot shell process specifically.

3

Storage Layer — World-Writable Dirs

```
find / -type d -perm -o+w 2>/dev/null
```

Finds directories anyone can write to — a known security risk. `/tmp` is typically world-writable by design. Flag it in your report!

4

Write Your Audit Report

```
echo '=== SECURITY AUDIT REPORT ===' > audit_report.txt
echo 'Date: Week 8 Lab Session' >> audit_report.txt
echo 'Total User Accounts:' >> audit_report.txt
wc -l /etc/passwd >> audit_report.txt
echo 'Total Running Processes:' >> audit_report.txt
ps aux | wc -l >> audit_report.txt
cat audit_report.txt
```

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105–115 MIN

❌ Common Pitfalls — Live Error Fix Demo

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These are the top mistakes students make in every session. If something's not working, check here first before asking the TA.

Pitfall	What You See	Fix
Wrong emulator image	Blank screen or 'Select image' page	Select **Buildroot** from the Images dropdown. NOT FreeDOS or Windows 98. Wait for `/ #` prompt.
Lost in wrong directory	Empty `ls -l` output	Run `pwd` to check location. Run `cd /home/security_lab` to navigate back.
Confusing `>` vs `>>`	File content disappears	`>` overwrites. `>>` appends. Single arrow = shoots old content away!
Wrong octal in chmod	Unexpected permissions	Formula: r=4, w=2, x=1. Write digit by digit: owner·group·others. Double-check with `ls -l`.
Colab cell won't run	Nothing happens on click	Click INSIDE the cell first, then press **Shift+Enter**. Check for Python syntax errors — mismatched quotes are common.
`ps aux` overwhelms	Too many lines to read	Use `ps aux | head -10` for first 10 lines, or pipe to `grep` to filter for specific processes.

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🔧 TA Cheat Sheet — Analogies That Land



Confidentiality → Aadhaar Card

Only you and UIDAI can see your full Aadhaar details. Ask students: "Can your neighbour check your Aadhaar balance online?" → No! That's confidentiality.



Integrity → IRCTC Train Ticket

Only the system can modify the departure time — not passengers. If anyone could edit the ticket, chaos! Integrity ensures only authorized changes happen.



Availability → UPI During Diwali

BHIM/UPI must stay up even during peak festival load. What if the app went down during Diwali shopping? That's an availability failure.



RBAC → IIT Jodhpur Portal

A student sees grades. A professor enters grades. Their role determines access — no overlap allowed. This is RBAC in your daily college life.

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⚡ TA Quick Command Reference

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Diagnostic Commands

```
# Check current user
whoami

# Show all files including hidden
ls -la

# Show file contents with line numbers
cat -n filename.txt

# Show last 5 lines of a file
tail -5 filename.txt

# Find a specific file by name
find / -name 'filename.txt' 2>/dev/null

# Show disk usage
df -h

# Show who is logged in
who
```

Recovery Commands

```
# If student deletes a file by accident
echo 'content' > filename.txt

# If chmod seems to not work — use full path
chmod 600 /home/security_lab/secret_data.txt

# If copy.sh/v86 freezes
# → Refresh browser, restart emulation (~60 sec)

# If Colab disconnects
# → Click 'Reconnect' (top right)
# → All code auto-saved in notebook

# Count instances of a running process
```

ⓘ Buildroot supports: `ls, cat, echo, mkdir, cd, chmod, ps, wc, grep, find, head, tail`. Stick to these!

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115–120 MIN

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🎟 Exit Ticket — Answer One Before You Leave

The TA will ask each student to answer **one** of the following questions verbally or in the chat. There are no wrong attempts — just honest reflections.

1

CIA Pillar

Name one CIA pillar and give a real-life Indian example of it — Aadhaar, IRCTC, UPI, your college portal...

2

chmod Decode

What does `chmod 644` mean in terms of read/write/execute for owner, group, and others?

3

Auth vs. Authz

What is the difference between **Authentication** and **Authorization**? Give a one-line example for each.

4

Process Audit

Which command would you use to find all running processes on a Linux system and filter for a specific one?

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📚 Homework — Due Before Next Session

1 Terminal Task — Permission Setup

Create `/home/iit_lab` in copy.sh/v86. Inside it, create four files: `student_grades.txt`, `admin_config.txt`, `shared_notes.txt`, and `setup.sh`. Apply appropriate permissions for each based on the scenarios discussed in class. Take a screenshot of `ls -l /home/iit_lab` output and submit.

2 Colab Extension — New File + Chart

Extend the CIA Triad Simulator from Activity 3. Add a 4th file: `exam_paper.txt` with `authorized_readers=['admin']` and `authorized_writers=['admin']`. Test at least 3 access scenarios and generate a new updated bar chart with all results.

3 Reflection Paragraph

Writing Prompt: In your IIT Jodhpur assignment submission portal, which access control model is being used — DAC, MAC, or RBAC? Justify your answer with specific examples from your own experience using the portal.

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Learning Objectives — Did We Hit Them All?

By the end of this session, you should be able to confidently do each of the following:



Explain CIA + AAA

Confidentiality, Integrity, Availability + Authentication, Authorization, Accountability — with real Indian examples.



Compare DAC, MAC, RBAC

Know who decides access, whether the owner can override, and give a real-world example for each model.



Read Permission Strings

Decode `-rwxr-xr--` character by character — file type, owner, group, others.



Use chmod & chown

Apply both symbolic and octal chmod. Know that `600=private`, `644=public read`, `700=private executable`.



Map Attack Surfaces

Use `ps aux`, `cat /etc/passwd`, and `find` to investigate the Identity, Process, and Storage layers.

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🌟 Today You Were Real Linux Admins!

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> Today you all acted like real Linux system administrators! You set file permissions, explored attack surfaces, and built a CIA Triad simulator in Python. Every command you typed today is used by security professionals in actual companies. Keep experimenting — **curiosity is your best security tool!**

— IIT Jodhpur | Operating Systems | Week 8 | OS Security & Permissions

🔒 Stay Curious

The best security analysts never stop asking "what if someone tried to break this?"

🛡️ Stay Secure

Apply least-privilege thinking everywhere — in code, in systems, and in life.

🚀 See You Week 9

Submit homework before next session. Questions? Post in the course forum.

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